

## Template: how a CGEL fragment is structured

Fragment 0 — 2026-05-07

A fragment in this series describes a CGEL phenomenon as a class of trees satisfying a formal specification, in the model-theoretic idiom of Rogers (1998), the GPSG tradition (Gazdar, Klein, Pullum & Sag 1985), and Pullum’s Edinburgh handouts. Each fragment is self-contained, two to four pages, and proceeds in roughly the same shape:

- (1) Phenomenon. A CGEL construction or constraint, with a worked English example and the CGEL-style tree.
- (2) Vocabulary. The unary and binary predicates needed beyond the shared ones in `shared/macros.tex`.
- (3) Specification. The defining formula(e) in FOL or wMSO on trees, with glosses.
- (4) Worked instance. The example from (1) shown to satisfy the specification.
- (5) Open questions. Where the formalism strains, what richer logic might be needed, what coordination-style residues (à la Gazdar 1981) remain.

## 1. Phenomenon

*Stub.* Replace with the construction and an example. Use `\ea/\z` for numbered examples:

- (1) *This*, I think she knew.

## 2. Vocabulary

*Stub.* List any new predicates beyond `shared/macros.tex`. For instance:

- (2)  $\text{Phrasal}(x)$ ,  $\text{NP}(x)$ ,  $\text{Head}(x, y)$ ,  $\text{Comp}(x, y)$ .

## 3. Specification

*Stub.* Define predicates with  $\equiv_{\text{def}}$ :

- (3)  $\text{Vacancy}(x) \equiv_{\text{def}} \text{Phrasal}(x) \wedge \neg(\exists y)[x \triangleleft y]$   
*a vacancy is a phrasal node with no children*

## 4. Worked instance

*Stub.* Show the example tree satisfies the specification.

## 5. Open questions

*Stub.* Coordination residue, gradience, lexical exceptions, adequacy of FOL vs. wMSO for the construction.